SafeHer Team Handover

SafeHer is a personal-safety system: Android app + FastAPI backend + on-device audio AI + WhatsApp/Firebase alerts + live track web page.

Repo: https://github.com/Ghige/SAFEHER

Folders: safeher_android / safeher_backend / safeher_ai / safeher_web. Large training audio (safeher_ai/data) and secrets are NOT in Git.

1. First-time setup (Backend) - start here

- Install Python 3.11 (recommended). Avoid 3.12+ for AI/TensorFlow compatibility.
- Install PostgreSQL and Redis (Redis optional for simple local API tests; required for reliable dead-man switch).
- Clone the repo from GitHub.
- Open a terminal in safeher_backend.
- Create venv: python -m venv .venv
- Activate (Windows): .venv\Scripts\activate
- Activate (macOS/Linux): source .venv/bin/activate
- Install deps: pip install -r requirements.txt
- Copy credentials template: copy .env.example .env (then fill real values - never commit .env)
- Place Firebase Admin JSON locally (e.g. firebase-credentials.json). Point FIREBASE_CREDENTIALS_PATH to it.
- Create DB, then: alembic upgrade head
- Run API: uvicorn app.main:app --reload --host 0.0.0.0 --port 8000
- Open docs: http://localhost:8000/docs

2. Put credentials (local only)

Ask the lead for private values. Do not put real secrets in Git, chat dumps, or this PDF.

- Backend .env: DATABASE_URL, SECRET_KEY, REDIS_URL, META_* WhatsApp tokens, FIREBASE_*, TRACK_BASE_URL, API_BASE_URL
- Backend Firebase Admin SDK JSON file (service account)
- Android: google-services.json into safeher_android/app/
- Android: local.properties with sdk.dir and MAPS_API_KEY
- Optional release: keystore.properties from keystore.properties.example

3. Android setup (short)

- Install Android Studio + JDK 17.
- Open safeher_android/, add google-services.json + local.properties.
- Sync Gradle, run debug app on device/emulator.
- Point app API base URL to your machine (emulator often 10.0.2.2:8000).
- TFLite model asset yamnet_distress_quant.tflite should already be under app assets.

4. AI training setup (optional / retrain)

- cd safeher_ai && python -m venv .venv && activate && pip install -r requirements_training.txt
- Dataset is NOT in Git. Get safeher_ai/data from shared Drive/OneDrive and place under safeher_ai/data/.
- Pipeline: training/00_convert_audio.py -> 01_prepare_dataset.py -> 02_train_model.py -> 03_evaluate_model.py -> 04_export_tflite.py
- Copy exported TFLite into Android assets before release.

5. Web track page

- safeher_web/track/index.html - contacts open WhatsApp track link with token.
- Locally backend can serve /track/. Production: host on safeher.app with query string preserved.

6. CRITICAL - Wallet / pocket panic auto-trigger (current behavior)

CURRENT STATUS: If the phone is in a wallet/pocket and the SafeHer app is in the BACKGROUND (screen off / app not open), AI does NOT keep listening for panic and will NOT auto-trigger SOS.

Why: AiMonitoringCoordinator only runs the inference loop while the app is in the foreground (_isAppInForeground). AppForegroundTracker stops the loop on background. Field docs also say background inference pauses. This is intentional for RC1 (battery/privacy/OS limits), but it means hands-free wallet detection is NOT done yet.

What DOES work today (when Monitoring is ON and app is open/foreground):

- On-device TFLite audio scoring about every 3 seconds
- Backend confidence fusion can recommend: ignore / haptic / 30s countdown / immediate SOS
- Manual SOS button + alert flow + WhatsApp track link + location while alert active
- Dead-man escalation if pings stop during an active alert (needs Redis in multi-worker prod)

7. Pending / remaining work (summary for team)

- AI: True always-on / wallet-pocket monitoring with reliable background service + battery policy
- AI: Reduce false positives (street noise, traffic); retune thresholds with real field data
- AI: Movement / fall model still feature-flagged OFF (AiFeatureFlags.MOVEMENT_AI_ENABLED = false)
- AI: Dataset sharing outside Git; improve distress/normal balance; optional noise folder for augmentation
- AI: Cloud dataset upload / Help Improve AI is limited (design: opt-in; not full prod pipeline)
- Messaging: Production Meta WhatsApp approved templates (alert + cancel) wired in .env
- Android: Official google-services.json, Maps key, release signing, battery-optimization UX polish
- Backend: Production Postgres/Redis/hosting, rotate any previously shared service-account keys
- Web: Deploy track page on production domain with CORS + HTTPS
- QA: Full end-to-end on real devices outdoors (see test cases below)
- Docs: Keep README + PRODUCTION_SETUP.md updated as env vars change

8. Test cases checklist

Backend

- GET /docs loads; health/auth register+login returns JWT
- pytest in safeher_backend passes
- Create alert -> WhatsApp/Meta path sends (or logs clearly if token missing)
- Track link opens map updates; cancel alert hides location
- Dead-man: stop pings -> escalation after timeout

Android - core

- Install, login/register, grant mic + location (+ notifications)
- Manual SOS -> countdown -> active alert -> contacts get track link
- Cancel SOS within window -> cancel message / no false emergency
- Maps show when MAPS_API_KEY set; graceful fallback if missing
- FCM token saved after login (users.fcm_token)

Android - AI (foreground)

- AI Status: on-device model Ready
- Monitoring ON: quiet room stays below trigger; no SOS from silence
- Normal talk should not repeatedly hit high distress
- Safe playback of distress sample via speaker raises score (note peak)
- Score tiers: ~60-79 countdown path; ~80+ immediate SOS path (see AiScoreThresholds / field doc)
- Monitoring OFF: mic foreground service stops

Android - wallet / background (expected FAIL until pending work ships)

- Enable Monitoring, lock phone / put in pocket/wallet for 2-5 minutes with distress audio nearby
- EXPECTATION TODAY: no auto SOS (inference paused in background) - confirm and log as known gap
- Bring app to foreground again - monitoring/inference should resume

Regression / safety

- No secrets committed (.env, adminsdk JSON, google-services.json)
- Release build ProGuard does not break Retrofit/Firebase
- Offline SOS queue behavior (if implemented on device) when network returns

9. Who to ask / where to read

- Root README.md - full setup
- PRODUCTION_SETUP.md - production checklist
- safeher_ai/README.md - training pipeline
- safeher_android/docs/AUDIO_AI_FIELD_TEST.md - AI field tests
- safeher_web/README.md - track page deploy

Never commit passwords, API tokens, Firebase private keys, or .env files.